

ORACLE DATABASE GUIDE

How to Drop a Parent Table with Active Foreign Key Constraints

1. Create the Parent Table (customers)

We start by creating the parent table `customers` which holds the primary key reference.

```
CREATE TABLE customers (  
    customer_id NUMBER PRIMARY KEY,  
    customer_name VARCHAR2(100) NOT NULL,  
    email VARCHAR2(100) UNIQUE  
);
```

```
SQL> CREATE TABLE customers (  
    customer_id NUMBER PRIMARY KEY,  
    customer_name VARCHAR2(100) NOT NULL,  
    email VARCHAR2(100) UNIQUE  
);
```

Table created.

```
SQL> desc customers;
```

Name	Null?	Type
CUSTOMER_ID	NOT NULL	NUMBER
CUSTOMER_NAME	NOT NULL	VARCHAR2(100)
EMAIL		VARCHAR2(100)

2. Create the Child Table (orders)

The `customer_id` column in `orders` acts as the foreign key, linking each order to a specific customer.

```
CREATE TABLE orders (  
    order_id NUMBER PRIMARY KEY,  
    order_date DATE DEFAULT SYSDATE,  
    total_amount NUMBER(10, 2),  
    customer_id NUMBER,  
    CONSTRAINT fk_customer  
        FOREIGN KEY (customer_id)  
        REFERENCES customers(customer_id)  
        ON DELETE CASCADE  
);
```

```
SQL> CREATE TABLE orders (  
    order_id NUMBER PRIMARY KEY,
```

```

        order_date DATE DEFAULT SYSDATE,
        total_amount NUMBER(10, 2),
        customer_id NUMBER,
        CONSTRAINT fk_customer
            FOREIGN KEY (customer_id)
            REFERENCES customers(customer_id)
            ON DELETE CASCADE
    );

```

Table created.

SQL> desc orders;

Name	Null?	Type
ORDER_ID	NOT NULL	NUMBER
ORDER_DATE		DATE
TOTAL_AMOUNT		NUMBER(10,2)
CUSTOMER_ID		NUMBER

Note: Adding ON DELETE CASCADE means if a customer record is deleted from the parent table, all of their related orders in the child table will automatically be deleted as well.

3. Insert Sample Data

Populating both parent and child tables with sample entries:

```

SQL> INSERT INTO customers (customer_id, customer_name, email) VALUES (1, 'Avijit',
'avijit@example.com');

```

1 row created.

```

SQL> INSERT INTO customers (customer_id, customer_name, email) VALUES (2,
'surojit', 'surojit@example.com');

```

1 row created.

```

SQL> COMMIT;

```

Commit complete.

```

SQL> SELECT * FROM customers;

```

CUSTOMER_ID	CUSTOMER_NAME	EMAIL
1	Avijit	avijit@example.com
2	Surojit	surojit@example.com

```

SQL> INSERT INTO orders (order_id, total_amount, customer_id) VALUES (5001, 249.99,
1);

```

1 row created.

```

SQL> INSERT INTO orders (order_id, total_amount, customer_id) VALUES (6001, 499.99,
2);

```

1 row created.

```

SQL> COMMIT;

```

Commit complete.

```
SQL> SELECT * FROM orders;
```

ORDER_ID	ORDER_DATE	TOTAL_AMOUNT	CUSTOMER_ID
5001	20-AUG-26	249.99	1
6001	20-AUG-26	499.99	2

4. How to Drop the Parent Table

If you attempt to run standard `DROP TABLE customers;` command, the database blocks the operation with an error (ORA-02449) because active foreign key constraints in orders reference it.

```
SQL> DROP TABLE CUSTOMERS;
```

```
DROP TABLE CUSTOMERS
```

```
*
```

```
ERROR at line 1:
```

```
ORA-02449: unique/primary keys in table referenced by foreign keys
```

To drop the parent table directly without needing to drop or alter the child table first, use the **CASCADE CONSTRAINTS** clause:

```
SQL> DROP TABLE customers CASCADE CONSTRAINTS;
```

```
Table dropped.
```

```
SQL> SELECT * FROM CUSTOMERS;
```

```
SELECT * FROM CUSTOMERS
```

```
*
```

```
ERROR at line 1:
```

```
ORA-00942: table or view does not exist
```